

Chu Cheng Qing Sec - 02

~~GA~~

- a. The ~~key~~ characteristics of Genetic Algorithms is ~~it uses selection, crossover, and mutation to evolve a population of candidate solutions over generations, while~~

Aspect	Genetic Algorithm (GA)	Particle Swarm Optimization (PSO)
Inspiration	Based on natural evolution and survival of the fittest	Based on social behavior of birds flocking or fish school schooling.
Population unit	Individual or chromosome	Particle
Solution representation	Usually encoded binary strings, real numbers, permutations or chromosomes	Usually represented as real-valued position vectors.

b:	P _{old}	G _{best}	P _{best}
	15	5	20
	2	8	8

$$P_v = (0, 1)$$

$$C_1 = 0.5$$

$$C_2 = 0.4$$

$$\text{rand}() = 0.3$$

$$\Delta t = 1.0$$

P 1st position

$$V_x = V_x + C_1 \times \text{rand}() \times (G_{best} - \text{Position}_x) + C_2 (2 \times \text{rand}()) \times (P_{best} - \text{Position}_x)$$

$$= 0 + [0.5 \times 0.3 \times (5 - 15)] + [0.4 \times 0.3 \times (20 - 15)]$$

$$= -1.5 + 0.6$$

$$V_x = -0.9$$

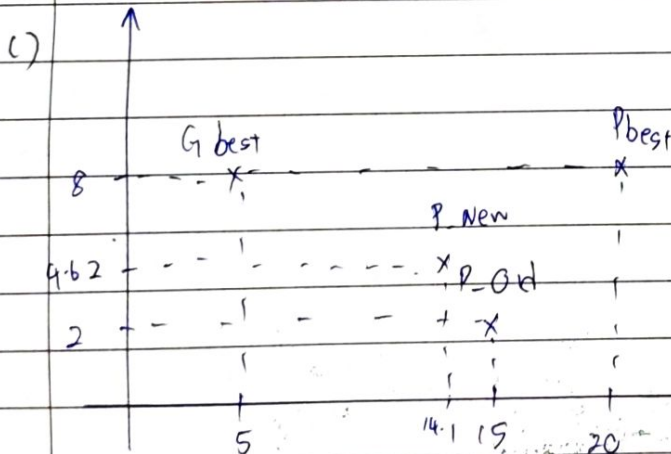
P second position

$$\begin{aligned}
 V_y &= V_y + [C_1 \times \text{rand}() \times (G_{\text{best}_y} - \text{Position}_y)] + [C_2 \times \text{rand}() \times (P_{\text{best}_y} - \text{Position}_y)] \\
 &= 1 + [0.5 \times 0.3 \times (8 - 2)] + [0.4 \times 0.3 \times (8 - 2)] \\
 &= 1 + 0.9 + 0.72 \\
 &= 2.62
 \end{aligned}$$

$$\begin{aligned}
 \text{Position}_x &= \text{Position}_x + (V_x \times \Delta t) \\
 &= 15 + [-0.9 \times 1.0] \\
 &= 14.1
 \end{aligned}$$

$$\begin{aligned}
 \text{Position}_y &= \cancel{2} + \cancel{[2.62]} \\
 &\quad \text{Position}_y + (V_y \times \Delta t) \\
 &= 2 + [2.62 \times 1.0] \\
 &= 4.62
 \end{aligned}$$

$P_v (14.1, 4.62)$



- d) The new position of particle P_i moves toward the global best position while still considering its own best position. PSO updates the particle direction by balancing both global and personal experience to search for a better solution.